Verdict Project Roadmap

Gridline — FIA penalty consistency tool | Personal reference — September 2026

Phase 1 — PDF Parsing

Status: In Progress (~80%)

Done:

- Extract text from FIA decision PDFs (pdfplumber)
- Split text into structured fields: Driver, Competitor, Session, Fact, Infringement, Decision, Reason
- Handle label wording variance across seasons (Infringement / Offence normalized to one field)
- Extract turn number, own car, other car, and penalty points via regex
- Fallback logic: pull "other car" from Reason when Fact doesn't mention a second car
- Batch-run the parser across a folder of PDFs, fault-tolerant per file
- Validated against four real, structurally different documents (grid drop + points, warning, no further action, older wording)

Remaining:

- Handle multi-driver "No / Driver" documents (rare, untested edge case)
- Persist parsed output somewhere durable (this is the immediate next step)

Phase 2 — Storage

Status: Not Started

Remaining:

- Save parsed results as JSON first (simple, inspectable, no schema commitment yet)
- Run the parser across a real, large batch of documents (dozens-hundreds, multiple seasons) to surface unseen template variants
- Once the field set feels stable, design a Postgres table and migrate, following the existing db/sql/ migration pattern

Phase 3 — Telemetry

Status: Barely Started

Done:

- Proof-of-concept script pulling FastF1 lap telemetry (speed, throttle, brake, position) for a known driver/lap

Remaining:

- Build the mapping from a parsed penalty record (race, driver, incident time) to the matching FastF1 session/lap
- Pull telemetry (speed, position, gap, who was ahead) for that exact moment
- This time-alignment step is a real unknown: FIA documents use wall-clock time, FastF1 telemetry is lap/session-relative

Phase 4 — Join

Status: Not Started

Remaining:

- Attach telemetry data to each penalty record in one combined structure or table
- Depends entirely on Phase 3's time-alignment work being solved first

Phase 5 — Comparison / Consistency Model

Status: Not Started

Remaining:

- Define what "similar incident" means (turn type, closing speed, contact severity, who was ahead, etc.)
- Build the actual logic that flags cases where similar incidents received different verdicts
- Most conceptually difficult phase of the project — an analysis/design problem, not just code

Phase 6 — Searchable UI

Status: Not Started

Remaining:

- Frontend to search and browse incidents, and surface flagged inconsistencies
- Likely extends the existing Gridline React app rather than a new one

Phase 7 — Productionizing

Status: Not Started

Remaining:

- Wire the parser into an Airflow DAG (matching existing DAGs) to run on a schedule
- Deploy alongside the rest of Gridline via the existing Docker Compose setup
- Handle new FIA documents automatically as each race weekend publishes them

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Reality Check

This is a multi-week project, not a few more sessions. Phase 1 (parsing one document type) is the easiest phase, and it's the only one solidly underway. The three hardest, most uncertain phases — telemetry-to-verdict time alignment, defining incident "similarity," and the comparison model itself — haven't been touched yet, and each is its own multi-step effort with genuine unknowns, not just functions to write.

Recommended Next Checkpoint

Not "production" — a proof point. Get 10–20 real parsed decisions saved as JSON, pull matching FastF1 telemetry for 2–3 of them, and get one incident fully joined end-to-end (PDF verdict + telemetry side by side). This proves the biggest technical unknown (time alignment) before investing further in parsing scale or the comparison model.